Announcements – Jan 26

- HW 2 is due next Friday at 9am.
- SI sessions are Tue 6-7pm and Thu 2-3pm.
- Quiz 5 trivia – Pong was the second arcade video game and the first that was broadly successful.

End of Section 1.4
Starting Section 1.5

Quantifiers with Multiple Variables

All the variables must be bound to make a predicate a proposition.

Examples:

$\forall x \, \exists y \, P(x,y)$ is a proposition (both x and y are bound)

$\exists x \, \exists y \, \exists z \, Q(x, y, z)$ is also a proposition (all vars bound)

$\forall x \; P(x,y)$ is a not proposition (y is free, so truth depends on value of y)

Combining Quantifiers

- **$\exists x \exists y$ P(x,y)** There exists an x and a y such that P(x,y) = T
- **$\exists x \forall y$ P(x,y)** There exists an x s.t. for every y P(x,y) = T
- **$\forall x \exists y$ P(x,y)** For every x there exists a y s.t. P(x,y) = T
- **$\forall x \forall y$ P(x,y)** For every x and for every y P(x,y) = T

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Suppose P(x,y) is x eats y, x$\in$ Mammals, y $\in$ Insects

Then $\exists \mathbf{x} \exists$y P(x,y) in plain English reads "There is a mammal that eats some insect."

$\exists \mathbf{x} \forall$y P(x,y) reads "Some mammal eats every insect."

$\forall \mathbf{x} \exists$y P(x,y) reads "Every mammal eats some insect"

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Can have even more: $\forall x \exists y \forall z (P(x, y) \rightarrow P(x,z))$

More Nested Quantifier Practice

Let $P(x, y)$ x teaches y

where the domain is UA Faculty × UA Courses

Write the expression for Dr. Anson teaches exactly 2 courses

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Let P(x, y) x teaches y

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Write the expression for Dr. Anson teaches exactly 2 courses

$$\exists w\, \exists z\, \forall y\, (P(\text{Anson}, w) \land P(\text{Anson}, z) \land (w \neq z) \land$$
$$(P(\text{Anson}, y) \rightarrow ((y = w) \lor (y = z)))$$

$w, z, y \in$ UA Courses

Mathematical Expressions

We can use quantified statements to express mathematical statements.

e.g. "The sum of two positive integers is always positive."

$\forall x \forall y \left(\big((x > 0) \land (y > 0)\big) \rightarrow (x + y > 0)\right)$

The domain is all integers.

"Every real number except 0 has a multiplicative inverse."

$\forall x \left((x \neq 0) \rightarrow \exists y (x \cdot y = 1)\right)$

The domain is all real numbers.

Varying the domain and predicates

Some student in this class owns a cat and a dog.

Translate this into a logical expression.

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Easiest

Domain is students in this class.

P(x) is owns a cat and a dog.

- $\exists$ x P(x)

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Translate this into a logical expression.

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Domain is students in this class.

P(x) is owns a cat and a dog.

- $\exists\, x\, P(x)$

Now let's require we use a different domain.

Varying the domain and predicates

Some student in this class owns a cat and a dog.

Translate this into a logical expression.

New domain, must contain all students in this class.

Domain is students at U of A.

P(x) is owns a cat and a dog.

C(x) x is in this class
$\exists x \, (C(x) \land P(x))$

Now add the requirement of using a predicate with two variables.

Varying the domain and predicates

Some student in this class owns a cat and a dog.

Translate this into a logical expression.

$P(x, y)$ is x owns y, $x \in$ Students at U of A, $y \in$ Animals

$C(x)$ x is in this class , $x \in$ Students at U of A

$\exists x\ (C(x) \land P(x,cat) \land P(x,dog))$

More Examples

Everybody loves somebody.

$\forall x \, \exists y \, P(x, y)$

P(x, y): x loves y

There is somebody who is loved by everyone.

$\exists y \, \forall x \, P(x, y)$

Order of quantifiers is important!

DeMorgan for Multiple Quantifiers

Recall:

$\neg\exists x\ P(x) \equiv \forall x\ \neg P(x)$

$\neg\forall x\ P(x) \equiv \exists x\ \neg P(x)$

This idea carries through for multiple quantifiers.

$\neg\exists x\ \forall y\ P(x, y) \equiv \forall x\ \neg\forall y\ P(x,y) \equiv \forall x\ \exists y\ \neg P(x,y)$

DeMorgan for Multiple Quantifiers

example: $P(x, y)$: x loves y Domain all people

$\forall x \, \exists y \, P(x, y)$ Everyone loves someone

$\neg \forall x \, \exists y \, P(x, y)$

$\exists x \, \forall y \, \neg P(x, y)$ There is someone who doesn't love anyone

Everyone Else

Suppose we have a domain of students in an art class. We want to express the sentence "Everyone in the class painted a portrait of someone else in the class."

We might try something like:

$M(x,y)$: x painted a portrait of y

$\forall x \, \exists y \, M(x, y)$

The problem is, this might evaluate to true, even if some student, Narcissus, only painted a self portrait. (i.e. M(Narcissus, Narcissus) is true, but M(Narcissus, y) is false for all other students y.)

How do we capture the "someone else" part of the proposition?

Everyone Else

Suppose we have a domain of students in an art class. We want to express the sentence "Everyone in the class painted a portrait of someone else in the class."

How about:

M(x,y): x painted a portrait of y

$\forall x \, \exists y \, ((x \neq y) \land M(x,y))$

This works.

Everyone Else

Suppose we have a domain of students in an art class. Now suppose we want to express the sentence "Everyone in the class painted a portrait of everyone else in the class." Notice that this can be true even if no one paints a self portrait.

As before, we can't do:

M(x,y): x painted a portrait of y

$\forall x \forall y\, M(x,y)$

Since everyone would have to paint a portrait of themselves for this to be true.

Everyone Else

Suppose we have a domain of students in an art class. Now suppose we want to express the sentence "Everyone in the class painted a portrait of everyone else in the class." Notice that this can be true even if no one paints a self portrait.

Last time we fixed our problem by adding the condition x is not equal to y:

M(x,y): x painted a portrait of y

$\forall x\ \forall y\ ((x \neq y) \land M(x, y))$

Does this fix our problem in this case?

No, this makes the statement false for almost all domains since every student equals themselves.

What is the one special case when the statement would be true?

Everyone Else

Suppose we have a domain of students in an art class. Now suppose we want to express the sentence "Everyone in the class painted a portrait of everyone else in the class." Notice that this can be true even if no one paints a self portrait.

We need an implication for this:

M(x,y): x painted a portrait of y

$\forall x \, \forall y \, ((x \neq y) \rightarrow M(x, y))$

Now it has to be true that every paints a portrait of everyone else?

There exists a unique

Recall $\exists! x\, P(x)$ means there's exactly one element in the domain that makes $P(x)$ true.

How can we express this idea in logic?

$\exists x\, \forall y\, (P(x) \wedge (P(y) \rightarrow (x = y))$

or

$\exists x\, \forall y\, (P(x) \wedge (x \neq y) \rightarrow \neg P(y))$

Moving Quantifiers

You may move a quantifier through a logical statement as long as you don't cross another quantifier or a variable of the same name.

For example, suppose we have the predicates:
$P(x)$: x is a secret agent
$D(y, x)$: y is the nemesis of x (The domain of both is all characters)

We can express the statement "every secret agent has a nemesis" as:

$\forall x\ (P(x) \rightarrow \exists y\ D(y, x))$

which is equivalent to

$\forall x\ \exists y\ (P(x) \rightarrow D(y, x))$

Moving Quantifiers

You may move a quantifier through a logical statement as long as you don't cross another quantifier or a variable of the same name. Also, if you cross a negation, you must use DeMorgan.

For example, suppose we have the predicates:
P(x): x is a secret agent
D(y, x): y is the nemesis of x

We can express the statement "every secret agent has a nemesis" as:

$\forall x \exists y (P(x) \to D(y, x))$

however this is not equivalent to

$\exists y \forall x (P(x) \to D(y, x))$

which means "there is someone who is the nemesis of all secret agents"

Moving Quantifiers

You actually can cross another quantifier, as long as it is of the same type.

P(x): x is a secret agent
D(y, x): y is the nemesis of x

We can express the statement "some secret agent has a nemesis" as:

$\exists x \, (P(x) \rightarrow \exists y \, D(y, x))$

this is equivalent to

$\exists y \, \exists x \, (P(x) \rightarrow D(y, x))$

which means "someone is the nemesis of some secret agent"
I've written the English a little differently, but they are logically the same.

Moving Quantifiers

You may move a quantifier through a logical statement as long as you don't cross another quantifier or a variable of the same name.

We cannot move across variables of the same name:

For example there is something that makes P(x) true and there is something that makes Q(x) true can be written as:

$\exists x\ P(x) \wedge \exists x\ Q(x)$

this is NOT the same as:

$\exists x\ \exists x\ (P(x) \wedge Q(x))$

which is a poorly written way of saying there is something that makes both P(x) and Q(x) true.

Prenux Normal Form (PNF)

A quantified statement is in Prenux Normal Form (PNF) if all the quantifiers appear first.

For example these are in PNF:

$\exists x \, \forall y \, (P(x) \land \, Q(y))$

$\exists x \exists y \forall z \, ((P(x, y) \lor \big(Q(z)\big) \rightarrow R(x, y, z))$

These are not in PNF:

∃x (P(x) → ∃y D(y, x))

$\exists x \; P(x) \land \exists x \; Q(x)$

For every quantified statement, there is an equivalent quantified statement that is in PNF.
Casually: “Every quantified statement can be written in PNF.”

Prenux Normal Form (PNF)

For every quantified statement, there is an equivalent quantified statement that is in PNF.
Casually: “Every quantified statement can be written in PNF.”

How?

If every quantifier has a different variable, then just push them all to the left without crossing any of the quantifiers.

$\forall x\ (P(x) \rightarrow \exists y\ D(y, x))$

$\forall x\ \exists y\ (P(x) \rightarrow D(y, x))$

Prenux Normal Form (PNF)

For every quantified statement, there is an equivalent quantified statement that is in PNF.
Casually: "Every quantified statement can be written in PNF."

What do we do if some names are the same?

We first change the variables names so that no quantifier has the same variable name

$\exists x\ P(x) \wedge \exists x\ Q(x)$

$\exists x\ P(x) \wedge \exists y\ Q(y)$

$\exists x \exists y\ (P(x) \wedge Q(y))$